Complete Derivations

Every Object and Variable in the Canonical–BSDT Framework

From First Principles to Final Formula

Contents

1 Primitives: State, Features, Metric

1.1 State matrix X

Definition 1.1 (State)**.** The state is a matrix

$$X \in \mathbb{R}^{N \times d}$$

where N is the number of agents and d is the feature dimension per agent. Row i is the state of agent i: $x_i \in \mathbb{R}^d$. In vectorised form: $\mathrm{vec}(X) \in \mathbb{R}^{Nd}$.

Dimensions.

Symbol	Meaning	Type
N	number of agents	positive integer
d	features per agent	positive integer
$n = Nd$	total state dimension	positive integer
X	full state	$\mathbb{R}^{N \times d}$
x_i	state of agent i	$\mathbb{R}^d$

1.2 Feature map S

Definition 1.2 (Feature map)**.**

$$S : \mathbb{R}^{N \times d} \to \mathbb{R}^k, \qquad X \mapsto S(X)$$

$S(X)$ encodes the geometric content relevant to the control objective. $k \geq 1$ is fixed per domain. The Jacobian of S at X is

$$J(X) = \frac{\partial S}{\partial X} \in \mathbb{R}^{k \times n}, \quad n = Nd.$$

Jacobian in index notation. Let $S = (S^1, \ldots, S^k)^\top$ and $X = (X^1, \ldots, X^n)^\top$ (vectorised). Then:

$$J_{ab}(X) = \frac{\partial S^a}{\partial X^b}, \quad a \in \{1, \ldots, k\},\ b \in \{1, \ldots, n\}.$$

Chain rule. For any trajectory $X(t)$:

$$\dot{S} = J(X)\dot{X} \in \mathbb{R}^k.$$

1.3 Metric G

Definition 1.3 (Feature-space metric)**.**

$$G \in \mathbb{R}^{k \times k}, \quad G = G^\top \succ 0.$$

Eigenvalue bounds: $0 < \mu_G \leq \lambda(G) \leq M_G$.

Derivation of eigenvalue bounds. Since $G \succ 0$, all eigenvalues are positive. Order them $0 < \lambda_1 \leq \cdots \leq \lambda_k$. Define:

$$\mu_G := \lambda_1 = \lambda_{\min}(G), \qquad M_G := \lambda_k = \lambda_{\max}(G).$$

For any $v \in \mathbb{R}^k$:

$$\mu_G \left\| v \right\|^2 \leq v^\top G v \leq M_G \left\| v \right\|^2.$$

These bounds are tight: achieved at the eigenvectors of G.

1.4 Derivation of μ_G and M_G for BSDT

Under the BSDT instantiation $G = \Sigma_0^{-1}$:

$$\mu_G = \lambda_{\min}(\Sigma_0^{-1}) = \frac{1}{\lambda_{\max}(\Sigma_0)}, \qquad M_G = \lambda_{\max}(\Sigma_0^{-1}) = \frac{1}{\lambda_{\min}(\Sigma_0)}.$$

Proof: if $\Sigma_0 v = \lambda v$ then $\Sigma_0^{-1} v = \lambda^{-1} v$, so eigenvalues invert. $\square$

Conditioning:

$$\kappa(G) = \frac{M_G}{\mu_G} = \frac{\lambda_{\max}(\Sigma_0)}{\lambda_{\min}(\Sigma_0)} = \kappa(\Sigma_0).$$

2 Energy E

2.1 Definition

$$\boxed{E(X) = S(X)^\top G\, S(X) \in \mathbb{R}_{\geq 0}}$$

2.2 Properties

Proposition 2.1 (Non-negativity and zero set). *$E(X) \geq 0$ with equality if and only if $G^{1/2} S(X) = 0$, i.e. $S(X) = 0$ (since $G \succ 0$).*

Proof. $E = S^\top G S = \left\| G^{1/2} S \right\|^2 \geq 0$. Equality holds iff $G^{1/2} S = 0$ iff $S = 0$ since $G^{1/2}$ is invertible. $\square$

2.3 Time derivative of E

Proposition 2.2 (Energy velocity). *Along any trajectory $X(t)$:*

$$\dot{E}(t) = \langle g_X(X(t)), \dot{X}(t) \rangle$$

where $g_X = \nabla_X E$ is derived in Section 3.

Proof. By the chain rule:

$$\dot{E} = \frac{d}{dt}[S^\top G S] = 2 S^\top G \dot{S} = 2 S^\top G J \dot{X} = (2 J^\top G S)^\top \dot{X} = g_X^\top \dot{X}.$$

$\square$

2.4 BSDT instantiation of E

With $S = \tilde{X} \Sigma_0^{-1/2}$ (vectorised), $G = I$:

$$E_{\mathrm{BSDT}} = \left\| \tilde{X} \Sigma_0^{-1/2} \right\|_F^2 = \operatorname{tr}(\tilde{X} \Sigma_0^{-1} \tilde{X}^\top)$$

where $\tilde{X} = X - \mathbf{1}\mu_0^\top$.

With $S = B(X) = (\delta_C, \delta_G, \delta_A, \delta_T)^\top$, $G = A$:

$$E_{\mathrm{BSDT}} = B(X)^\top A\, B(X) = E_{\mathrm{BS}}(S).$$

3 State-Space Gradient g_X

3.1 Definition

$$g_X(X) = \nabla_X E(X) = 2J(X)^\top G\, S(X) \in \mathbb{R}^n$$

3.2 Derivation from first principles

Proof. Write $E = \sum_{a,b} G_{ab} S^a S^b$. Differentiate with respect to X^j:

$$\frac{\partial E}{\partial X^j} = \sum_{a,b} G_{ab} \left(\frac{\partial S^a}{\partial X^j} S^b + S^a \frac{\partial S^b}{\partial X^j} \right) = 2 \sum_{a,b} G_{ab} J_{aj} S^b = 2[J^\top GS]_j.$$

Hence $\nabla_X E = 2J^\top GS$. □

3.3 Matrix form

In matrix notation with X as an $N \times d$ matrix (not vectorised), the gradient is an $N \times d$ matrix:

$$\nabla_X E = 2J^\top GS \in \mathbb{R}^{N\times d}$$

where $J \in \mathbb{R}^{k\times Nd}$ acts on $\mathrm{vec}(\dot{X})$ and the result is reshaped.

3.4 Norm of g_X

Proposition 3.1 (Gradient norm)**.**

$$\|g_X\|^2 = 4S^\top GJJ^\top GS = 4S^\top G\,\mathcal{G}\,GS$$

where $\mathcal{G} = JJ^\top \in \mathbb{R}^{k\times k}$ is the Gram matrix of J.

Proof. $\|g_X\|^2 = (2J^\top GS)^\top(2J^\top GS) = 4S^\top GJJ^\top GS$. □

3.5 Coercivity bounds on $\|g_X\|$

Proposition 3.2 (Two-sided gradient coercivity)**.** *Under (A1)–(A2) with $\sigma_{\min}(J) \geq \sigma > 0$ and $\|J\|_{\mathrm{op}} \leq J_{\max}$:*

$$C_1\sqrt{E} \leq \|g_X\| \leq C_2\sqrt{E}$$

where

$$C_1 = \frac{2\sigma\mu_G}{\sqrt{M_G}}, \qquad C_2 = \frac{2J_{\max}M_G}{\sqrt{\mu_G}}.$$

Proof. **Lower bound.**

$$\|g_X\|^2 = 4\left\|J^\top GS\right\|^2 \geq 4\sigma_{\min}(J)^2\|GS\|^2 \geq 4\sigma^2\mu_G^2\|S\|^2.$$

Since $E = S^\top GS \leq M_G\|S\|^2$, we have $\|S\|^2 \geq E/M_G$. Therefore $\|g_X\|^2 \geq 4\sigma^2\mu_G^2 E/M_G = C_1^2 E$.

Upper bound.

$$\|g_X\| = 2\left\|J^\top GS\right\| \leq 2\|J\|_{\mathrm{op}}\|G\|_{\mathrm{op}}\|S\| \leq 2J_{\max}M_G\|S\|.$$

Since $E = S^\top GS \geq \mu_G\|S\|^2$, we have $\|S\| \leq \sqrt{E/\mu_G}$. Therefore $\|g_X\| \leq 2J_{\max}M_G\sqrt{E/\mu_G} = C_2\sqrt{E}$. □ □

3.6 BSDT pullback gradient $\tilde{G}_t$

In the BSDT system, the gradient decomposes per channel:

$$g_X = 2 \sum_{k \in \{C,G,A,T\}} [AS]_k \frac{\partial \delta_k}{\partial X} =: \tilde{G}_t.$$

This is the pullback gradient of Section XVI.3 of the anatomy document. The canonical formula $g_X = 2J^\top GS$ recovers this exactly under the identification $G = A$, $S = B(X)$, $J = \partial B / \partial X$.

4 MFLS: Mahalanobis Field Line Score

4.1 BSDT definition

$$\mathrm{MFLS}_{\mathrm{BSDT}}(X) = \|\nabla_X E_{\mathrm{BSDT}}\|_F = 2 \left\| \tilde{X} \Sigma_0^{-1} \right\|_F$$

4.2 Canonical derivation

$$\boxed{\mathrm{MFLS}(X) = \|g_X\| = 2 \left\| J^\top GS \right\|}$$

Proof. By definition $\mathrm{MFLS} = \|\nabla_X E\|_F$. The canonical gradient is $\nabla_X E = g_X = 2J^\top GS$. Therefore $\mathrm{MFLS} = \left\|2J^\top GS\right\| = \|g_X\|$. □

4.3 Expanded forms

Three equivalent canonical expressions:

Form	Expression	When to use
Gradient norm	$\|g_X\|$	Free — already computed in canonical loop
Explicit	$2\left\|J^\top GS\right\|$	When gradient vector is needed
Gram form	$2\sqrt{S^\top G \mathcal{G} GS}$	When Gram structure is analysed

4.4 MFLS squared

$$\mathrm{MFLS}^2 = \|g_X\|^2 = 4S^\top G \mathcal{G} GS = 4S^\top GJJ^\top GS$$

4.5 MFLS as Fisher information

Under $G = I_S$ (Fisher metric):

$$\mathrm{MFLS}^2 = 4\mathrm{tr}(\Sigma_0^{-1} \tilde{X}^\top \tilde{X} \Sigma_0^{-1}) = 4\mathrm{tr}(I_{\mathrm{Fisher}})$$

MFLS squared is four times the trace of the Fisher information matrix at the current state. High MFLS means the system is in a region of high log-likelihood curvature.

4.6 MFLS as leading indicator (proved)

From the energy derivative with $F_{\text{base}} = 0$:

$$\dot{E} = -(1-\gamma)\left\|g_X\right\|^2 = -(1-\gamma)\cdot \text{MFLS}^2.$$

MFLS squared is the instantaneous energy descent rate. A spike in MFLS precedes a drop in E. This is a theorem, not an empirical observation.

4.7 Verification against BSDT

Under $S = B(X)$, $G = \Sigma_0^{-1}$, $J = \partial B/\partial X$ with $J_C = 2\tilde{X}\Sigma_0^{-1}$ (the camouflage Jacobian):

$$g_X = 2J_C^{\mathsf{T}}\Sigma_0^{-1}S = 2\tilde{X}\Sigma_0^{-1}, \qquad \|g_X\|_F = 2\left\|\tilde{X}\Sigma_0^{-1}\right\|_F = \text{MFLS}_{\text{BSDT}}. \checkmark$$

5 Adaptive Gain γ

5.1 Definition

$$\boxed{\gamma(X) = \frac{E(X)}{E(X)+\theta}, \qquad \theta > 0}$$

5.2 Derivation from the saddle-point game

Consider the two-player zero-sum game at fixed X:

- Controller chooses $\gamma \in [0,1)$
- Adversary chooses $u \in \mathbb{R}^n$ with $\|u\| \leq M$
- Payoff to adversary: $J(\gamma, u) = \dot{E}(X;\gamma,u)$

The payoff under the canonical ODE is:

$$J(\gamma,u) = (1-\gamma)\left[\langle g_X,\, F_{\text{base}} + u\rangle - \left\|g_X\right\|^2\right].$$

The adversary maximises: by Cauchy–Schwarz, $u^* = M_{g_X}/\left\|g_X\right\|$.

The controller minimises:

$$\inf_{\gamma\in[0,1)}(1-\gamma)B, \quad B = \langle g_X,\, F_{\text{base}}\rangle + M_{\|g_X\| - \|g_X\|^2}.$$

When $B > 0$: infimum is 0 as $\gamma \to 1^-$; boundary solution. When $B \leq 0$: minimum at $\gamma = 0$.

The Lagrangian formulation (Theorem O.1 of v4):

$$\mathcal{L}(u,\lambda) = \tfrac{1}{2}\left\|u - F_{\text{base}}\right\|^2 + \lambda\langle u,\, g_X\rangle - \frac{\theta}{2}\frac{\left\|g_X\right\|^2}{E}\lambda^2$$

has unique saddle point with $\lambda^* = \frac{\langle F_{\text{base}}, g_X, E\rangle}{(E+\theta)\|g_X\|^2}$, giving $\gamma = E/(E+\theta)$.

5.3 Properties

Proposition 5.1 (Gain properties). *(i) $\gamma \in [0, 1)$ for all X*

(ii) $\gamma \to 0$ as $E \to 0$: no damping near target

(iii) $\gamma \to 1$ as $E \to \infty$: full projection at high energy

(iv) $\gamma = 1/2$ when $E = \theta$: maximum-entropy operating point

(v) $\gamma' = \theta/(E + \theta)^2 > 0$: strictly increasing in E

Proof. (i) $E \geq 0$, $\theta > 0$, so $0 \leq E/(E+\theta) < 1$. (ii) $\lim_{E\to 0} E/(E+\theta) = 0$. (iii) $\lim_{E\to\infty} E/(E+\theta) = 1$. (iv) $E/(E+\theta) = 1/2 \iff E = \theta$. (v) $d\gamma/dE = \theta/(E+\theta)^2 > 0$. □

5.4 Fermi-Dirac structure

Writing $\gamma = 1/(1+\theta/E)$: this is the Fermi-Dirac occupation function with inverse temperature $\beta = 1/\theta$ and energy E. The thermodynamic dictionary (v4 §N):

Canonical object	Thermodynamic analogue
θ	Temperature
γ	Occupation number
$1 - \gamma$	Carnot efficiency
$E = \theta$	Maximum-entropy operating point
$\theta \to 0$	Zero temperature (full intervention)
$\theta \to \infty$	Infinite temperature (no intervention)

5.5 Formula for θ

$$\theta = \text{median}_{t\in\text{calm}}\{E(t)\} = \text{median}_{t\in\text{calm}}\left\{\left\|\tilde{X}_t\Sigma_0^{-1/2}\right\|_F^2\right\}$$

Derivation. The median of the calm-period energy distribution is the natural temperature because:

1. At $E = \theta$, $\gamma = 1/2$ — half intervention. During normal operation ($E \approx \theta$ by construction), the system applies moderate damping.
2. The median is robust to outlier spikes in the calibration window. The mean would overstate θ.
3. Setting $\theta = E_{\text{median}}$ means the system is at its maximum-entropy operating point on average during the calm regime — it neither over-intervenes nor under-intervenes.

Alternative derivations of θ:

Route 1 (Maximum entropy): set $\theta = e^*$ (critical manifold energy) so $\gamma = 1/2$ at stability boundary.

Route 2 (Admissibility): choose $\theta > M_{\max}\sqrt{M_G}/(\sigma\mu_G)$ to guarantee $\Psi^* > M_{\max}$.

Route 3 (Chi-squared calibration): $e^* = \chi^2_{Nd,0.99}$, set $\theta = e^*/2$ so $\gamma = 1/3$ at the 99% energy quantile.

6 The Canonical ODE

6.1 Locked form

$$\boxed{\dot{X} = F_{\text{base}}(X) - g_X(X) - \gamma(X)\,\frac{\langle F_{\text{base}}(X) - g_X(X),\, g_X(X)\rangle}{\|g_X(X)\|^2}\,g_X(X)}$$

6.2 Decomposition

Define the modified force:

$$F(X) = F_{\text{base}}(X) - g_X(X).$$

The projection coefficient:

$$\alpha(X) = \frac{\langle F(X),\, g_X(X)\rangle}{\|g_X(X)\|^2}.$$

The ODE becomes:

$$\dot{X} = F(X) - \gamma(X)\alpha(X)g_X(X).$$

Three-term decomposition:

1. $F_{\text{base}}(X)$: domain-specific nominal field
2. $-g_X(X)$: gradient correction forcing E to decrease
3. $-\gamma\alpha g_X$: adaptive projection removing the component of F aligned with g_X

6.3 Force decomposition

$$\dot{X} = \underbrace{F - \langle F, \hat{u}\rangle\hat{u}}_{F_\perp \text{ (always free)}} + \underbrace{(1-\gamma)\langle F, \hat{u}\rangle\hat{u}}_{(1-\gamma)F_\parallel \text{ (damped)}}$$

where $\hat{u} = g_X / \|g_X\|$.

At $E \to 0$: $\gamma \to 0$, $\dot{X} \to F_{\text{base}}$ (free dynamics). At $E \to \infty$: $\gamma \to 1$, $\dot{X} \to F_\perp$ (only safe motion).

7 Lyapunov Energy Derivative $\dot{E}$

7.1 Master identity

Theorem 7.1 (Energy derivative, Lemma 7.7 of v4)**.** *Along any solution of the canonical ODE:*

$$\dot{E} = (1-\gamma)\left[\langle g_X,\, F_{\text{base}}\rangle - \|g_X\|^2\right]$$

Proof.

$$\begin{aligned}
\dot{E} &= \langle g_X, \dot{X} \rangle \\
&= \langle g_X, F_{\text{base}} - g_X - \gamma\alpha g_X \rangle \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma\alpha \|g_X\|^2 \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma\langle F, g_X \rangle \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma\langle F_{\text{base}} - g_X, g_X \rangle \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma\langle F_{\text{base}}, g_X \rangle + \gamma \|g_X\|^2 \\
&= (1-\gamma)\langle g_X, F_{\text{base}} \rangle - (1-\gamma) \|g_X\|^2 \\
&= (1-\gamma) \left[\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 \right]. \quad \square
\end{aligned}$$

□

7.2 Special cases

Pure geometric flow ($F_{\text{base}} = 0$):

$$\dot{E} = -(1-\gamma) \|g_X\|^2 = -(1-\gamma) \cdot \text{MFLS}^2 \leq 0.$$

Strictly negative whenever $g_X \neq 0$ and $\gamma < 1$.

Descent condition (general F_{base}):

$$\dot{E} \leq 0 \iff \langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2 .$$

At the maximum-entropy point ($E = \theta$, $\gamma = 1/2$):

$$\dot{E} = \tfrac{1}{2} \left[\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 \right].$$

8 Effective Decay Rate ρ_{eff}

8.1 Definition

$$\rho_{\text{eff}}(t) = -\frac{\dot{E}(t)}{E(t)}$$

8.2 Derivation in canonical terms

From the energy derivative:

$$\rho_{\text{eff}} = \frac{(1-\gamma) \left[\|g_X\|^2 - \langle g_X, F_{\text{base}} \rangle \right]}{E}.$$

Under $F_{\text{base}} = 0$:

$$\rho_{\text{eff}} = \frac{(1-\gamma) \|g_X\|^2}{E} = \frac{(1-\gamma) \cdot \text{MFLS}^2}{E}.$$

8.3 Connection to σ

From the coercivity bound $\|g_X\|^2 \geq C_1^2 E$:

$$\rho_{\text{eff}} \geq (1-\gamma)C_1^2 = (1-\gamma)\frac{4\sigma^2\mu_G^2}{M_G} = \rho_{\text{theoretical}}.$$

The theoretical rate ρ is the worst-case lower bound. The effective rate ρ_{eff} is the instantaneous observed rate. The ratio $\rho_{\text{eff}}/\rho \geq 1$ measures how much better the system is performing relative to the theoretical guarantee.

8.4 Half-life prediction

The local exponential half-life is:

$$\tau_{1/2}(t) = \frac{\ln 2}{\rho_{\text{eff}}(t)}.$$

This is the time at which E will reach $E(t)/2$ at the current decay rate.

9 Effective Singular Value σ_{eff}

9.1 Abstract definition

$\sigma = \sigma_{\min}(J(X))$ from Theorem 9.3 of v4 is the abstract worst-case bound.

9.2 Derivation in canonical terms

$$\boxed{\sigma_{\text{eff}}(t) = \frac{\|g_X(t)\|\sqrt{M_G}}{2\mu_G\sqrt{E(t)}} = \frac{\text{MFLS}_t\sqrt{M_G}}{4\mu_G\sqrt{E_t}}}$$

Proof. From the coercivity bound:

$$\|g_X\|^2 \geq \frac{4\sigma^2\mu_G^2}{M_G}E.$$

Treat this as an equality to define the effective σ that is tight at the current state:

$$\|g_X\|^2 = \frac{4\sigma_{\text{eff}}^2\mu_G^2}{M_G}E \implies \sigma_{\text{eff}}^2 = \frac{\|g_X\|^2 M_G}{4\mu_G^2 E} \implies \sigma_{\text{eff}} = \frac{\|g_X\|\sqrt{M_G}}{2\mu_G\sqrt{E}}.$$

Since MFLS $= \|g_X\|$ (Section 4):

$$\sigma_{\text{eff}} = \frac{\text{MFLS}\sqrt{M_G}}{2\mu_G\sqrt{E}}.$$

□

9.3 Properties

$\sigma_{\text{eff}}(t) \geq \sigma$ always (effective is at least as good as worst case).

At the coercivity lower bound: $\sigma_{\text{eff}} = \sigma$ (bound is tight when J is at its worst orientation).

σ_{eff} is a time-varying signal measuring the current Jacobian rank quality — computable at each step.

10 Theoretical Rate ρ

10.1 Definition

$$\rho = \frac{4\sigma^2\mu_G^2}{M_G}(1-\gamma_{\max}), \qquad \gamma_{\max} = \frac{E(X_0)}{E(X_0)+\theta}$$

10.2 Derivation

From Theorem 9.3 of v4 with $F_{\text{base}} = 0$:

$$\dot{E} = -(1-\gamma)\left\|g_X\right\|^2.$$

Apply the coercivity lower bound:

$$\left\|g_X\right\|^2 \geq \frac{4\sigma^2\mu_G^2}{M_G}E.$$

Since $\gamma \leq \gamma_{\max} = E_0/(E_0+\theta) < 1$ (monotone in E, decreasing, so maximum at $t=0$):

$$\dot{E} \leq -(1-\gamma_{\max})\frac{4\sigma^2\mu_G^2}{M_G}E = -\rho E.$$

By Grönwall: $E(t) \leq E_0 e^{-\rho t}$.

10.3 Expression via σ_{eff}

Substituting $\sigma = \sigma_{\text{eff}}$:

$$\rho_{\text{eff}} = \frac{4\sigma_{\text{eff}}^2\mu_G^2}{M_G}(1-\gamma) = \frac{4\cdot\frac{\|g_X\|^2 M_G}{4\mu_G^2 E}\cdot\mu_G^2}{M_G}(1-\gamma) = \frac{\left\|g_X\right\|^2}{E}(1-\gamma).\checkmark$$

Consistent with the direct definition of ρ_{eff}.

11 Admissibility Threshold Ψ^*

11.1 Abstract form

From Theorem 9.4 of v4:

$$\Psi^* = \frac{\sigma\theta\mu_G}{\sqrt{M_G}}.$$

This is unimodal with maximum at $R^* = \theta$.

11.2 Derivation in canonical terms

$$\Psi^* = \frac{\theta\left\|g_X\right\|}{4\sqrt{E}} = \frac{\theta\cdot\text{MFLS}}{4\sqrt{E}}$$

Proof. Substitute $\sigma = \sigma_{\text{eff}} = \|g_X\| \sqrt{M_G}/(2\mu_G\sqrt{E})$:

$$\Psi^* = \frac{\sigma_{\text{eff}}\theta\mu_G}{\sqrt{M_G}} = \frac{\|g_X\| \sqrt{M_G}}{2\mu_G\sqrt{E}} \cdot \frac{\theta\mu_G}{\sqrt{M_G}} = \frac{\theta \|g_X\|}{2\sqrt{E}}.$$

Since $\text{MFLS} = \|g_X\|$:

$$\Psi^* = \frac{\theta \cdot \text{MFLS}}{2\sqrt{E}}.$$

□

Note on the factor. The formula above uses σ_{eff} (instantaneous effective singular value). Using the abstract worst-case σ gives $\Psi^*_{\text{abstract}} = \sigma\theta\mu_G/\sqrt{M_G}$, while the canonical instantaneous formula gives $\Psi^*(t) = \theta\text{MFLS}_t/(2\sqrt{E_t})$. Both are valid; the canonical form is computable at each step.

11.3 BSDT instantiation

With $G = \Sigma_0^{-1}$, $\mu_G = 1/\lambda_{\max}(\Sigma_0)$, $M_G = 1/\lambda_{\min}(\Sigma_0)$:

$$\Psi^*_{\text{BSDT}} = \frac{\sigma\theta/\lambda_{\max}(\Sigma_0)}{\sqrt{1/\lambda_{\min}(\Sigma_0)}} = \frac{\sigma\theta\sqrt{\lambda_{\min}(\Sigma_0)}}{\lambda_{\max}(\Sigma_0)} = \frac{\sigma\theta}{\sqrt{\kappa(\Sigma_0)} \cdot \lambda_{\max}^{1/2}(\Sigma_0)}.$$

The canonical form $\theta\text{MFLS}/(2\sqrt{E})$ is metric-free — the Σ_0 terms cancel. □

11.4 Sensitivity to calibration conditioning

From the BSDT formula:

$$\Psi^*_{\text{BSDT}} \propto \frac{1}{\sqrt{\kappa(\Sigma_0)}}.$$

A 10× increase in calibration conditioning shrinks Ψ^* by $\sqrt{10} \approx 3.16\times$. This is the gap closure result G7.2.

12 Admissibility Condition

12.1 Statement

$$\boxed{\Psi^* > M_{\max} \iff \frac{\theta \cdot \text{MFLS}_t}{2\sqrt{E_t}} > M_{\max} \iff \frac{\text{MFLS}_t}{\sqrt{E_t}} > \frac{2M_{\max}}{\theta}}$$

12.2 Derivation

Starting from the abstract condition:

$$\Psi^* > M_{\max}$$

Substituting the canonical form:

$$\frac{\theta \cdot \text{MFLS}}{2\sqrt{E}} > M_{\max}$$

Rearranging:

$$\text{MFLS} > \frac{2M_{\max}\sqrt{E}}{\theta}$$

Dividing both sides by $\sqrt{E}$:

$$\frac{\text{MFLS}}{\sqrt{E}} > \frac{2M_{\max}}{\theta}.$$

12.3 The canonical admissibility ratio

$$\boxed{\mathcal{A}(t) = \frac{\text{MFLS}_t}{\sqrt{E_t}} = \frac{\|g_X(t)\|}{\sqrt{E(t)}}}$$

This ratio is the canonical admissibility signal. It measures gradient strength relative to energy level.

The admissibility condition is then simply:

$$\mathcal{A}(t) > \frac{2M_{\max}}{\theta}.$$

Connection to ρ_{eff}:

$$\rho_{\text{eff}}(t) = \mathcal{A}(t)^2(1 - \gamma(t)).$$

A system decaying fast is a system that is safe. Fast decay implies admissibility.

13 Safety Ratio

13.1 Definition

$$\boxed{\text{SafetyRatio}(t) = \frac{\Psi^*(t)}{M_{\max}} = \frac{\theta \cdot \text{MFLS}_t}{2M_{\max}\sqrt{E_t}}}$$

13.2 Derivation

Divide the admissibility condition by $M_{\max}$:

$$\text{SafetyRatio} = \frac{\Psi^*}{M_{\max}} = \frac{\theta\text{MFLS}}{2M_{\max}\sqrt{E}}.$$

Substituting $\text{MFLS} = \|g_X\|$, E, θ — all already computed in the canonical loop:

Zero additional computation.

13.3 Regimes

SafetyRatio	Regime	Interpretation
> 1	Admissible	$\Psi^* > M_{\max}$; ultimate boundedness guaranteed
$= 1$	Boundary	Exactly at the admissibility threshold
< 1	Inadmissible	Disturbance can overcome canonical brake
$\to 0$	Critical	$\text{MFLS} \to 0$ or $E \to \infty$; system losing gradient signal

13.4 Connection to effective decay rate

$$\text{SafetyRatio}(t) = \frac{\theta}{2M_{\max}}\mathcal{A}(t) = \frac{\theta}{2M_{\max}}\sqrt{\frac{\rho_{\text{eff}}(t)}{1-\gamma(t)}}.$$

This connects the safety guarantee directly to the observable performance metric.

14 Alignment Angle $\cos\theta_t$

14.1 Definition

$$\boxed{\cos\theta_t = \frac{\langle F_{\text{base}}(X_t),\, g_X(X_t)\rangle}{\|F_{\text{base}}(X_t)\| \cdot \|g_X(X_t)\|} \in [-1,+1]}$$

14.2 Derivation from energy descent

The descent condition $\dot{E} \le 0$ requires:

$$\langle g_X,\, F_{\text{base}}\rangle \le \|g_X\|^2 .$$

Writing $\langle g_X,\, F_{\text{base}}\rangle = \|g_X\|\,\|F_{\text{base}}\|\cos\theta_t$:

$$\|F_{\text{base}}\|\cos\theta_t \le \|g_X\| .$$

When $\cos\theta_t \le 0$: descent is automatic regardless of $\|F_{\text{base}}\|$. When $\cos\theta_t > 0$: descent requires $\|F_{\text{base}}\|\cos\theta_t < \|g_X\|$, i.e. F_{base} is not pushing into the energy gradient too strongly.

14.3 Energy derivative in terms of $\cos\theta_t$

$$\dot{E} = (1-\gamma)\,\|g_X\|\left[\|F_{\text{base}}\|\cos\theta_t - \|g_X\|\right].$$

14.4 Geometric interpretation

Value	Meaning
$\cos\theta_t \to +1$	F_{base} co-aligned with g_X — coherent collapse pressure
$\cos\theta_t \approx 0$	F_{base} tangent to energy level set — no radial progress
$\cos\theta_t \to -1$	Anti-aligned — restoring competition, E falls without F_{base} help

14.5 Collapse angle $\tan\theta_t$

From §XIV of the anatomy document:

$$\tan\theta_t = \frac{\|F_\perp\|}{\|F_\parallel\|} = \sqrt{\frac{1}{\cos^2\theta_t} - 1} = \frac{\sqrt{1-\cos^2\theta_t}}{\cos\theta_t}.$$

$\tan\theta_t \to 0$: trajectory fully collapse-directed. $\tan\theta_t \to \infty$: motion fully perpendicular, locally safe. $\tan\theta_t = 1$ ($\theta_t = 45$): critical balance.

15 Misalignment Angle ψ_t (Sixth Signal)

15.1 Definition

$$\cos\psi_t = \frac{\langle \tilde{G}_t,\, g_t \rangle_F}{\left\|\tilde{G}_t\right\|_F \cdot \|g_t\|}$$

where $g_t = \nabla_S E_{\mathrm{BS}}(S_t) \in \mathbb{R}^4$ is the channel-space gradient and $\tilde{G}_t = J_t^{\mathsf{T}} g_t \in \mathbb{R}^{Nd}$ is its state-space pullback.

15.2 Derivation

The BSDT master operator uses g_t (channel space) in the projection. The canonical ODE uses $\tilde{G}_t$ (state-space pullback).

These differ by the Jacobian map: $\tilde{G}_t = J_t^{\mathsf{T}} g_t$.

The angle between them:

$$\cos\psi_t = \frac{\langle \tilde{G}_t, g_t, \rangle F}{\left\|\tilde{G}_t\right\|_F \|g_t\|} = \frac{R_t}{\left\|\tilde{G}_t\right\|_F \|g_t\|}$$

where $R_t = \sum_k [g_t]_k \langle \partial_X \delta_k, g_t \rangle_F$ is the control coupling sum.

15.3 What ψ_t measures

$\psi_t \approx 0$: channel-space risk and physical dynamics pointing in the same direction. Collapse is real. All five prior signals are reliable.

$\psi_t \approx \pi/2$: complete misalignment. Abstract risk elevated but physical dynamics not aligned with collapse. This is a false alarm.

15.4 False alarm resolution

Previously: high MFLS but no actual collapse. Unexplained by any single-space signal.

Now: high channel MFLS ($\|g_t\|$ large) with ψ_t near $\pi/2$ means $\tilde{G}_t \perp g_t$ — physical dynamics are tangential to the collapse direction. The alarm is real in channel space but not in state space. ψ_t discriminates between the two.

15.5 Performance gap formula

The performance difference between BSDT and CEK-v4:

$$\Delta\text{performance}(t) \approx f(\psi_t) + \kappa(\mathcal{R}_t)^{-1}$$

where:

- ψ_t large: BSDT projects using wrong direction, canonical corrects
- $\kappa(\mathcal{R}_t)$ large: channels nearly collinear, canonical compression is more robust

16 Channel Recovery Operator $\mathcal{R}$

16.1 The recovery formula

$$S_t = \mathcal{R}(X_t)\, g_X(t), \qquad \mathcal{R}(X) = \frac{1}{2} A^{-1} (JJ^\top)^{-1} J \in \mathbb{R}^{k\times n}$$

16.2 Derivation

From $g_X = 2J^\top GS$, left-multiply by J:

$$Jg_X = 2JJ^\top GS.$$

Since $JJ^\top \in \mathbb{R}^{k\times k}$ is invertible under the rank condition:

$$GS = \frac{1}{2}(JJ^\top)^{-1} J g_X.$$

Since $G = A \succ 0$ is invertible:

$$S = \frac{1}{2} G^{-1} (JJ^\top)^{-1} J g_X = \frac{1}{2} A^{-1} (JJ^\top)^{-1} J g_X = \mathcal{R} g_X.$$

16.3 Per-channel recovery

$$[\delta_k] = [S]_k = [\mathcal{R} g_X]_k, \quad k \in \{C, G, A, T\}.$$

16.4 Attribution

$$a_k(t) = \frac{\tilde{S}_k(t)^2}{\left\|\tilde{S}_t\right\|^2}, \quad \tilde{S}_k = \frac{S_k}{\mu_k^{\text{normal}}}, \quad \sum_k a_k = 1.$$

16.5 Stability of recovery

Proposition 16.1 (Lipschitz stability). *For perturbed gradient $\tilde{g_X} = g_X + \varepsilon$ with $\|\varepsilon\| \le \delta$:*

$$\left\|\tilde{S} - S\right\| \le \|\mathcal{R}\|_{\text{op}} \cdot \delta = \frac{\delta}{2\sigma_{\min}(A) \cdot \sigma_{\min}(JJ^\top)^{1/2} \cdot \sigma_{\min}(J)}.$$

Proof. $\tilde{S} - S = \mathcal{R}\varepsilon$, so $\left\|\tilde{S} - S\right\| \le \|\mathcal{R}\|_{\text{op}} \|\varepsilon\|$. Expanding $\|\mathcal{R}\|_{\text{op}} \le \frac{1}{2}\left\|A^{-1}\right\|_{\text{op}} \left\|(JJ^\top)^{-1}\right\|_{\text{op}} \|J\|_{\text{op}}$ and substituting singular value bounds gives the result. □

16.6 Condition number of recovery

$$\kappa(\mathcal{R}) = \frac{\sigma_{\max}(\mathcal{R})}{\sigma_{\min}(\mathcal{R})}.$$

Recovery is well-conditioned when $\kappa(\mathcal{R}) \ll 1$.

Recovery fails (ill-conditioned) when:

- Channels nearly collinear: $\sigma_{\min}(JJ^\top) \approx 0$
- Coupling matrix ill-conditioned: $\kappa(A) \gg 1$
- Jacobian rank-deficient: $\sigma_{\min}(J) \approx 0$ (channel has zero sensitivity)

17 Ultimate Boundedness Threshold Ψ^*_{rob}

17.1 Standard case

From Theorem 9.4 of v4:

$$\Psi_{\text{std}}(R) = (1 - \gamma(R)) \cdot 2\sigma\sqrt{\frac{\mu_G^2}{M_G}R} = \frac{\theta}{R+\theta} \cdot C_1\sqrt{R}.$$

Setting $u = \sqrt{R}$:

$$\Psi_{\text{std}}(u) = \frac{\theta C_1 u}{u^2 + \theta}.$$

Maximum at $u = \sqrt{\theta}$ (i.e. $R = \theta$):

$$\Psi^*_{\text{std}} = \frac{C_1\sqrt{\theta}}{2}.$$

17.2 Robust case with adversarial disturbance

With two-sided coercivity $C_1\sqrt{E} \le \|g_X\| \le C_2\sqrt{E}$:

$$\Psi_{\text{rob}}(R) = \frac{\theta C_1\sqrt{R}}{R + M_{C2}\sqrt{R} + \theta} - M.$$

Setting $u = \sqrt{R}$:

$$\frac{d\Psi_{\text{rob}}}{du} = \frac{\theta C_1(\theta - u^2)}{(u^2 + M_{C2}u + \theta)^2}.$$

Positive for $u < \sqrt{\theta}$, negative for $u > \sqrt{\theta}$: unique maximum at $u = \sqrt{\theta}$, i.e. $R = \theta$.

Unimodality proved.

17.3 Admissibility condition (robust)

$$M_< \Psi^*_{\text{rob}} = \Psi_{\text{rob}}(\theta) = \frac{\theta C_1\sqrt{\theta}}{\theta + M_{C2}\sqrt{\theta} + \theta} - M.$$

18 The Complete Dashboard

18.1 All signals from two objects

Every signal in the canonical dashboard is a function of g_X and E:

Signal	Formula	Objects	Section
MFLS_t	$\|g_X\|$	g_X	§4
E_t	$S^\mathsf{T} G S$	S, G	§2
γ_t	$E/(E+\theta)$	E, θ	§5
$\dot{E}_t$	$(1-\gamma)(\langle g_X,\, F_{\text{base}}\rangle - \|g_X\|^2)$	$g_X, E, \theta, F_{\text{base}}$	§7
$\rho_{\text{eff}}(t)$	$\|g_X\|^2\,(1-\gamma)/E$	g_X, E, θ	§8
$\cos\theta_t$	$\langle F_{\text{base}}, g_X,\, /\rangle(\|F_{\text{base}}\|\,\|g_X\|)$	g_X, F_{base}	§11
$\tan\theta_t$	$\sqrt{1/\cos^2\theta_t - 1}$	$\cos\theta_t$	§11
$\sigma_{\text{eff}}(t)$	$\|g_X\|\,\sqrt{M_G}/(2\mu_G\sqrt{E})$	g_X, E, μ_G, M_G	§9
$\Psi^*(t)$	$\theta\,\|g_X\|\,/(2\sqrt{E})$	g_X, E, θ	§10
$\mathcal{A}(t)$	$\|g_X\|\,/\sqrt{E}$	g_X, E	§10
$\mathrm{SafetyRatio}(t)$	$\theta\,\|g_X\|\,/(2M_{\max}\sqrt{E})$	g_X, E, θ	§10
$\tau_{1/2}(t)$	$\ln 2/\rho_{\text{eff}}(t)$	ρ_{eff}	§8
ψ_t	$\arccos(R_t/(\left\|\tilde{G}_t\right\|\,\|g_t\|))$	J, g_t, g_X	§12
S_t (channels)	$\mathcal{R}(X_t) g_X(t)$	J, A, g_X	§13
$a_k(t)$	$\tilde{S}_k^2/\left\|\tilde{S}\right\|^2$	S_t	§13

18.2 Dependency graph

```
X
 S = B(X)           E = S^T G S

                     = E/(E+)
                    _max = E_0/(E_0+)
                     = 4²²_G/M_G · (1-_max)

 J = S/X          g_X = 2J^T G S = MFLS

                      MFLS = ||g_X||
                      _eff = MFLS·M_G / (2_G·E)
                      * = ·MFLS / (2E)
                      SafetyRatio = */M_max
                      _eff = MFLS²·(1-)/E
                      cos _t = <F_base, g_X> / (||F_base||·MFLS)
                      R = R(X)·g_X → channels S_t → a_k(t)
```

18.3 Computational algorithm (per step t)

1. Compute $\tilde{X}_t = X_t - \mathbf{1}\mu_0^\mathsf{T}$
2. Compute $S_t = B(X_t) = (\delta_C, \delta_G, \delta_A, \delta_T)^\mathsf{T}$
3. Compute $J_t = \partial B/\partial X|_{X_t}$

4. Compute $g_X(t) = 2J_t^\mathsf{T} G S_t$ *(canonical gradient)*
5. Compute $E_t = S_t^\mathsf{T} G S_t$ *(energy)*
6. Compute $\gamma_t = E_t/(E_t + \theta)$ *(gain)*
7. Compute $\mathrm{MFLS}_t = \|g_X(t)\|$ *(free from step 4)*
8. Compute $\rho_{\text{eff}}(t) = \mathrm{MFLS}_t^2(1 - \gamma_t)/E_t$
9. Compute $\Psi^*(t) = \theta \mathrm{MFLS}_t/(2\sqrt{E_t})$
10. Compute $\mathrm{SafetyRatio}(t) = \Psi^*(t)/M_{\max}$
11. Compute $\cos\theta_t = \langle F_{\text{base}}(X_t),\, g_X(t)\rangle/(\|F_{\text{base}}\|\, \mathrm{MFLS}_t)$
12. Compute $\mathcal{R}_t = \frac{1}{2} A^{-1}(J_t J_t^\mathsf{T})^{-1} J_t$ *(4×4 inversion)*
13. Compute $S_t = \mathcal{R}_t g_X(t)$ *(channel recovery)*
14. Compute $a_k(t) = \tilde{S}_k(t)^2 / \left\|\tilde{S}_t\right\|^2$ *(attribution)*
15. Compute $\psi_t = \arccos(R_t/(\left\|\tilde{G}_t\right\|_F \|g_t\|))$
16. Update ODE: $\dot{X} = F_{\text{base}} - g_X - \gamma_t \alpha_t g_X$

19 BSDT–Canonical Juxtaposition

19.1 Identity table

Object	BSDT form	Canonical form	Equal?
State	$X \in \mathbb{R}^{N\times d}$	$X \in \mathbb{R}^{N\times d}$	Identical
Energy	$e_t = \left\|\tilde{X}\Sigma_0^{-1/2}\right\|_F^2$	$E = S^\mathsf{T} G S$	Equal under $G = A$
Gradient	$\tilde{G}_t = \sum_k [AS]_k \partial_X \delta_k$	$g_X = 2J^\mathsf{T} G S$	Identical
Gain	$\gamma^* = e_t/(e_t + \theta)$	$\gamma = E/(E + \theta)$	Identical
ODE	$\dot{X} = F_t - \gamma^* \frac{\langle F_t, g_t, \rangle}{\|g_t\|^2} g_t$	locked canonical ODE	Identical
MFLS	$2\left\|\tilde{X}\Sigma_0^{-1}\right\|_F$	$\|g_X\|$	Identical
Rate ρ	not proved	$4\sigma^2\mu_G^2/M_G(1-\gamma_{\max})$	canonical proves BSDT
Ψ^*	not proved	$\theta\mathrm{MFLS}/(2\sqrt{E})$	canonical proves BSDT
Channels	$(\delta_C, \delta_G, \delta_A, \delta_T)$ explicit	recovered via $\mathcal{R} g_X$	4×4 inversion
ψ_t	not in BSDT	$\arccos(R_t/\left\|\tilde{G}_t\right\| \|g_t\|)$	new in canonical
Extensions	none	7 extensions	canonical adds

19.2 The master identity

$$\underbrace{S_t}_{\text{BSDT channels}} = \underbrace{\frac{1}{2} A^{-1}(J_t J_t^\mathsf{T})^{-1} J_t}_{\text{recovery operator } \mathcal{R}(X_t)} \underbrace{g_X(t)}_{\text{canonical gradient}}$$

BSDT is on the left. Canonical is on the right. The recovery operator in the middle is the bridge. The equation says they are the same object, read from different directions.

19.3 Performance gap

BSDT uses g_t (channel space) in the projection. Canonical uses $\tilde{G}_t$ (state-space pullback). The gap is ψ_t:

$$\Delta\text{performance}(t) \sim \psi_t + \kappa(\mathcal{R}_t)^{-1}.$$

The v34→v58 empirical progression was systematically reducing ψ_t and $\kappa(\mathcal{R}_t)$ without knowing it:

Version	Implicit improvement	Canonical equivalent
v34	Base	No channel recovery
v48	4-quadrant modulator	Partial recovery via $(G_{\text{mem}}, T_{\text{mem}})$
v52	Clip structure	Excludes high $\kappa(\mathcal{R})$ regimes
v55	State-dependent $\theta_G(t)$	Ricci-flow metric adaptation
v58	Asymmetric clamp	Ill-conditioned recovery region excluded

20 Summary Reference

20.1 Every formula

$S = B(X) = (\delta_C, \delta_G, \delta_A, \delta_T)^\mathsf{T}$	feature vector
$E = S^\mathsf{T} G S$	Lyapunov energy
$J = \partial S / \partial X \in \mathbb{R}^{k \times n}$	Jacobian
$g_X = 2 J^\mathsf{T} G S$	canonical gradient
$\mathrm{MFLS} = \lVert g_X \rVert = 2 \left\lVert J^\mathsf{T} G S \right\rVert$	field line score
$\gamma = \dfrac{E}{E + \theta}$	adaptive gain
$\theta = \mathrm{median}_{t \in \mathrm{calm}} \{E(t)\}$	intervention cost
$\dot{E} = (1 - \gamma) \left[\langle g_X, F_{\mathrm{base}} \rangle - \lVert g_X \rVert^2 \right]$	energy derivative
$\rho_{\mathrm{eff}} = \dfrac{\lVert g_X \rVert^2 (1 - \gamma)}{E} = \dfrac{\mathrm{MFLS}^2 (1 - \gamma)}{E}$	instantaneous rate
$\sigma_{\mathrm{eff}} = \dfrac{\mathrm{MFLS} \sqrt{M_G}}{2 \mu_G \sqrt{E}}$	effective singular value
$\rho = \dfrac{4 \sigma^2 \mu_G^2}{M_G} (1 - \gamma_{\max})$	theoretical rate
$\Psi^* = \dfrac{\theta \cdot \mathrm{MFLS}}{2 \sqrt{E}}$	admissibility threshold
$\mathcal{A} = \dfrac{\mathrm{MFLS}}{\sqrt{E}} = \dfrac{\lVert g_X \rVert}{\sqrt{E}}$	admissibility ratio
$\mathrm{SafetyRatio} = \dfrac{\theta \cdot \mathrm{MFLS}}{2 M_{\max} \sqrt{E}}$	real-time safety signal
$\cos \theta_t = \dfrac{\langle F_{\mathrm{base}}, g_X \rangle}{\lVert F_{\mathrm{base}} \rVert \, \lVert g_X \rVert}$	alignment
$\cos \psi_t = \dfrac{\langle \tilde{G}_t, g_t \rangle_F}{\left\lVert \tilde{G}_t \right\rVert_F \lVert g_t \rVert}$	misalignment (sixth signal)
$\mathcal{R} = \dfrac{1}{2} A^{-1} (J J^\mathsf{T})^{-1} J$	channel recovery operator
$S_t = \mathcal{R}(X_t) g_X(t)$	channel recovery formula
$a_k = \tilde{S}_k^2 / \left\lVert \tilde{S} \right\rVert^2$	channel attribution

20.2 The two-object reduction

Everything derives from g_X and E.

g_X carries: MFLS, σ_{eff}, Ψ^*, SafetyRatio, $\cos \theta_t$, channel recovery.

E carries: γ, ρ_{eff}, $\tau_{1/2}$, admissibility condition.

The entire framework is a set of scalar and matrix operations on two vectors computed once per step.